

NIYTRI TECHNOLOGIES

AI Investment Intelligence Platform

Explainable, agentic AI for Indian markets — inside the broking experience.

Investor & Client Briefing

Version 1.1 · Built for SEBI-regulated broking

Confidential — for the intended recipient only. This document is AI-assisted and must be reviewed and approved before business or regulatory use.

1. Executive summary

The AI Investment Intelligence Platform is a production-ready software layer that brings explainable, governed artificial intelligence directly into an Indian broking application. It combines a multilingual conversational assistant, a daily agentic stock-scoring engine, market-news intelligence and portfolio analytics — all grounded in live market data and the firm's own research, and all wrapped in the governance a SEBI-regulated broker requires.

Unlike generic AI tools, every output is source-cited, every score is independently reviewed before publishing, and every action is captured in an immutable audit trail. The platform is vendor-neutral across leading AI providers, deploys on standard AWS infrastructure, and is engineered to scale with licensed data feeds and new channels.

2. The opportunity

India's retail investing base has expanded several-fold in recent years to well over 190 million demat accounts. Millions are first-time investors who need clear, trustworthy guidance rather than jargon or black-box tips — and they expect it instantly, in their own language, around the clock. Brokers that turn raw market data into compliant, conversational, explainable guidance deepen customer trust and engagement without adding headcount.

The problem today

- Information overload: prices, filings, news and ratings are scattered and overwhelming.
- No explainable guidance: customers ask "why?" and receive generic tips or silence.
- Compliance is heavy: every customer-facing word is regulated and manual review does not scale.
- Vendor & model risk: single-LLM tools break, drift, and lock the firm in with no governance trail.

3. Product overview

The platform delivers six customer-facing capabilities from a single, governed intelligence layer:

AI Assistant	Conversational, multilingual answers grounded in live quotes, AI scores, news and the firm's research — with source attribution and confidence.
Agentic Stock Scoring	A daily 8-agent pipeline scores every script 0–100 with plain-language drivers.
Market News Intelligence	Continuous collection, AI summaries, sentiment tags and impacted-stock mapping.
Portfolio Intelligence	Upload holdings (CSV/Excel) or connect via the back office; get a health score, concentration and sector analysis.
Personal Watchlist	Live price and AI score for the scripts each customer follows.
Dashboard	KPIs, score trends and top movers at a glance.

4. What sets it apart

An assistant that shows its work

Answers are retrieved (RAG) from live quotes, approved AI scores, today's news and the firm's research notes, then cited back to the customer. Hard compliance guardrails prevent any buy/sell/hold advice, keep the scoring methodology confidential, and keep the assistant focused on Indian equity markets.

Agentic scoring you can explain

A daily eight-agent pipeline blends eight factors — fundamentals, technicals, valuation, momentum, earnings, news sentiment, institutional activity and risk — into a 0–100 composite, and writes a concise rationale for every score.

Governance built in, not bolted on

- Maker-checker: an optional strict mode holds every score as pending until a human approves it.
- Independent AI checker: a second model reviews every rationale for compliance and accuracy before publishing.
- Full audit trail: every LLM call, decision, login and chat message is logged immutably.
- RBAC and chat audit: admin-defined roles with page-level access, and a searchable record of who asked what, when.

Multi-LLM engine, your control

Anthropic Claude, OpenAI GPT and Google Gemini sit behind one router with admin-configurable priority, per-provider model selection, and failover or round-robin strategies — no single-vendor dependency, with per-call usage and INR cost tracking.

5. Architecture

Frontend	React single-page app — responsive, light/dark, role-aware navigation.
Backend	Python FastAPI — REST APIs, the agentic pipeline, scheduler and governance.
Database	PostgreSQL on AWS RDS — users, scores, portfolios, audit and research vectors.
Intelligence	Multi-LLM router plus retrieval (RAG) over the firm's research documents.
Market data	Broker feeds → NSE → Yahoo fallback chain; RSS news; IST-aware scheduler.

6. Security & compliance

The platform is designed for a SEBI-regulated broker and is decision-support only — it never executes trades or gives personalised advice.

- AI disclaimer on every output; no buy/sell/hold recommendations anywhere.
- Scoring methodology kept confidential in the UI and assistant.

- JWT authentication with role-based access control; HTTPS end-to-end; secrets isolated from the codebase.
- Immutable audit log and a human maker-checker step before any score reaches customers.

7. Deployment & operations

- AWS-ready: one-command deployment on EC2 with the database on RDS and auto-renewing HTTPS certificates.
- Continuous integration runs build and tests on every change before production.
- Self-healing schema and an IST-scheduled daily scoring run with automatic catch-up.
- Live Agents dashboard, run history and INR cost tracking for full observability.

8. Roadmap

- Now (live): conversational AI, agentic scoring, governance, portfolio upload and multi-LLM routing.
- Next: fundamentals, earnings and FII/DII data feeds to enrich the neutral scoring pillars.
- Next: licensed broker market-data APIs for full, fast, production-grade coverage.
- Later: streaming responses, voice, and WhatsApp delivery.

9. About NIYTRI Technologies

NIYTRI Technologies designs, builds and deploys production AI software for regulated industries. We partner end-to-end — from product design through delivery and cloud deployment — to bring explainable, governed AI to your customers.